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Abstract

Individual tree detection (ITD) from canopy height models (CHM) derived via airborne LiDAR scanning occupies a central position in precision forest inventory workflows. Fixed-window local maxima (FWLM) approaches suffer from a well-known structural mismatch: a single search radius simultaneously over-detects broad crowns and misses narrow ones in structurally heterogeneous canopies. This systematic review examines variable and adaptive window methods for ITD, and, for the first time, frames them within the theory of spatial point processes—specifically the inhomogeneous Poisson process (NHPP), the log-Gaussian Cox process (LGCP), and Gibbs hard-core models—as a unified inferential platform for model calibration, residual diagnosis, and uncertainty quantification. The search spanned nine databases (Scopus, Web of Science, IEEE Xplore, ScienceDirect, SpringerLink, MDPI, Taylor & Francis, Google Scholar, PubMed) from January 2000 to May 2025, yielding 2,614 initial records that were screened following PRISMA 2020 guidelines down to 53 primary studies. Key findings encompass: (i) the evolution of variable radius functions from linear regression to semivariogram- and machine-learning-based formulations; (ii) the role of Ripley's K, L, g, and G statistics in diagnosing residual spatial structure in detected apex patterns; (iii) reported F1 scores ranging from 0.62 to 0.93 across forest types and sensor platforms; and (iv) the near-total absence of evidence for tropical montane and high-Andean forest environments. The combination of adaptive detectors with point-process modelling provides a promising avenue for automated calibration and rigorous cross-method comparison

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Adaptive Window Models for LiDAR-Based Individual Tree Detection: A Systematic Review from a Spatial Point Process Perspective and Its Challenges in High-Andean Ecosystems

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Abstract. Individual tree detection (ITD) from canopy height models (CHM) derived via airborne LiDAR scanning occupies a central position in precision forest inventory workflows. Fixed-window local maxima (FWLM) approaches suffer from a well-known structural mismatch: a single search radius simultaneously over-detects broad crowns and misses narrow ones in structurally heterogeneous canopies. This systematic review examines variable and adaptive window methods for ITD, and, for the first time, frames them within the theory of spatial point processes—specifically, the inhomogeneous Poisson process (NHPP), the log-Gaussian Cox process (LGCP), and Gibbs hard-core models—as a unified inferential platform for model calibration, residual diagnosis, and uncertainty quantification. The search spanned nine databases (Scopus, Web of Science, IEEE Xplore, ScienceDirect, SpringerLink, MDPI, Taylor & Francis, Google Scholar, PubMed) from January 2000 to May 2025, yielding 2,614 initial records that were screened following PRISMA 2020 guidelines down to 53 primary studies. Key findings encompass: *(i)* the evolution of variable radius functions from linear regression to semivariogram- and machine-learning-based formulations; *(ii)* the role of Ripley's K, L, g, and G statistics in diagnosing residual spatial structure in detected apex patterns; *(iii)* reported F_1 scores ranging from 0.62 to 0.93 across forest types and sensor platforms; and *(iv)* the near-total absence of evidence for tropical montane and high-Andean forest environments. The combination of adaptive detectors with point-process modelling provides a promising avenue for automated calibration and rigorous cross-method comparison.

Keywords: Individual tree detection · Adaptive local maxima · Canopy height model · Airborne LiDAR · Spatial point processes · Inhomogeneous Poisson process · Gibbs process · Ripley's K function · PRISMA systematic review

1 Introduction

Individual tree detection (ITD) constitutes the foundational step of remote-sensing-driven forest inventory pipelines [1,3,9]. Crown apex positions enable stand-level estimation of basal area, above-ground biomass, stem density, and structural diversity—all critical inputs for sustainable forest management and carbon-cycle monitoring [30,31,32].

Airborne Laser Scanning (ALS) is currently the most widely adopted sensor for ITD due to its capacity to yield high-density point clouds from which the Digital Surface Model (DSM), the Digital Terrain Model (DTM), and, by subtraction, the Canopy Height Model (CHM) can be derived [25,26]. The standard detection primitive applied to the CHM is the local maxima filter: a raster cell $\mathbf{s}^*$ is declared an apex when its height exceeds that of every neighbour within a moving window of radius r [1,3]. The fundamental weakness of a fixed radius is its inability to accommodate structural diversity within mixed-species or mixed-size stands: a single value of r will inevitably fragment large crowns into spurious multi-apex detections or merge adjacent narrow crowns into a single detection [8,10].

Variable Window Function (VWF) methods [2,**?**,13] address this by letting r depend on the local CHM height, spectral structure, or texture. Complementary approaches include watershed segmentation [5], region-growing algorithms [6], Gaussian mixture models [7], and, more recently, deep object detectors [19,**?**]. Nevertheless, rigorous cross-method comparison remains elusive because of inconsistent evaluation criteria, heterogeneous benchmark datasets, and divergent forest types under study [10,11].

One dimension of the ITD problem that has received disproportionately little attention is the use of *spatial point process theory* [34,35,36] as a principled framework for inference. Under this view, each set of detected apex locations constitutes a realization of a stochastic mechanism whose first-order intensity $\lambda(\mathbf{s})$ encodes the expected apex density per unit area as a function of environmental covariates [40,46]. Second-order summary statistics—Ripley's K and L functions [33], the pair correlation function $g(r)$, and the nearest-neighbour function $G(r)$—then provide diagnostic tools to determine whether residual structure in the detected pattern arises from genuine canopy ecology, detector artefacts, or both [46,34].

This article contributes a systematic review that: *(i)* synthesises the state of the art in adaptive window methods for ITD; *(ii)* analyses the spatial distribution of detected apices through a point-process lens; *(iii)* inventories the most commonly used metrics and reference datasets; and *(iv)* identifies research gaps for tropical montane and high-Andean forests. The review covers the period 2000–2025 and follows the PRISMA 2020 statement [47,48].

2 Theoretical Background

2.1 From LiDAR Point Cloud to CHM

CHM construction involves four sequential steps: (1) ground-point classification using Progressive TIN Densification [29] or the cloth simulation filter; (2) height normalisation by subtracting the interpolated DTM; (3) rasterisation via the pit-free algorithm of [27], which fills voids caused by laser penetration between branches; and (4) Gaussian or median smoothing to suppress high-frequency noise while preserving crown peaks. The entire workflow can be executed within the `lidR` package for R [28].

2.2 Fixed-Window Local Maxima (FWLM)

A pixel $\mathbf{s}^*$ is declared a local maximum under a fixed window of radius r_0 if:

$$Z(\mathbf{s}^*) = \max_{\mathbf{s}\in B(\mathbf{s}^*,\, r_0)} Z(\mathbf{s}), \tag{1}$$

where $Z(\mathbf{s})$ denotes the CHM height at location $\mathbf{s}$ and $B(\mathbf{s}^*, r_0)$ is the disc of radius r_0 centred at $\mathbf{s}^*$. Empirical studies on Virginia pine stands reported peak accuracy for $r_0 \in [2, 5]\,\mathrm{m}$ [1], yet transferability to other forest types remains limited [10].

2.3 Variable Window Function (VWF)

The natural extension of (1) is to parametrise the search radius as a function of the local CHM height:

$$r_{\text{lin}}(\mathbf{s}) = a + b\,Z(\mathbf{s}), \tag{2}$$

with $a > 0$ and $b \geq 0$ fitted by regression against field inventory data [2,13]. Power-law ($r = aZ^b$) and quadratic variants have been proposed for stands exhibiting nonlinear crown-to-height allometry [12,8]. A survey of 27 studies by [9] confirmed the linear VWF as the dominant benchmark in the literature, albeit with site-specific calibration that limits portability.

2.4 Structure-Driven Adaptive Window

Beyond height, the optimal window width should reflect canopy spatial heterogeneity at the local scale. [14] were among the first to derive the search radius from semivariograms computed within moving windows over IKONOS imagery, reporting F_1 gains of up to 12 % over the linear VWF in *Eucalyptus* plantations. Subsequent work has explored Moran's spatial autocorrelation [15], Haralick texture features [16], and Markov random fields [18] to guide window adaptation.

The most recent direction replaces handcrafted heuristics with learned predictors: [20] trained a Faster R-CNN detector on combined RGB and CHM imagery using weak LiDAR labels, resulting in the `DeepForest` package [21]. Separately, [6] proposed direct segmentation on the 3-D point cloud, bypassing the CHM rasterisation step and its associated information loss.

2.5 Spatial Point Process Models

A spatial point process Φ on $\mathcal{D} \subset \mathbb{R}^2$ is a random variable taking values in locally finite point sets [34,35]. Its first-order intensity is:

$$\lambda(\mathbf{s}) = \lim_{|\mathrm{d}\mathbf{s}| \to 0} \frac{\mathbb{E}[\Phi(\mathrm{d}\mathbf{s})]}{|\mathrm{d}\mathbf{s}|}, \tag{3}$$

encoding the expected apex rate per unit area.

Inhomogeneous Poisson Process (NHPP). Under this model, point locations are mutually independent and $\Phi(A) \sim \text{Poisson}\big(\int_A \lambda(\mathbf{s})\mathrm{d}\mathbf{s}\big)$ for any $A \subseteq \mathcal{D}$ [36]. The log-linear intensity model reads:

$$\log \lambda(\mathbf{s}; \boldsymbol{\beta}) = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}(\mathbf{s}), \tag{4}$$

where $\mathbf{x}(\mathbf{s})$ collects CHM-derived covariates such as height, local semivariogram, and terrain slope.

Log-Gaussian Cox Process (LGCP). When $\log \lambda(\mathbf{s}) = \mu(\mathbf{s}) + \xi(\mathbf{s})$ and ξ is a stationary Gaussian random field, the LGCP of [39] arises. This model captures over-dispersion produced by species-level canopy clustering.

Gibbs (Strauss) Process. To model physical inhibition between crowns—two trees cannot coexist closer than their combined crown radius—the Strauss model [38,37] is used:

$$f(\mathbf{x}) \propto \beta^{n(\mathbf{x})}\, \gamma^{s(\mathbf{x},R)}, \quad 0 \leq \gamma \leq 1, \tag{5}$$

where $n(\mathbf{x})$ is the number of detected points, $s(\mathbf{x}, R)$ counts pairs within distance R, and $\gamma = 0$ yields a hard-core process. [42] applied Gibbs models to mixed European stands, demonstrating that short-range inhibition is the rule in closed canopies.

Second-Order Summary Statistics. Ripley's K-function [33] is estimated by:

$$\widehat{K}(r) = \frac{|\mathcal{D}|}{n(n-1)} \sum_{i \neq j} \mathbf{1}(\|\mathbf{x}_i - \mathbf{x}_j\| \leq r)\, w_{ij}^{-1}, \tag{6}$$

with Ripley's edge correction w_{ij}^{-1}. Under complete spatial randomness (CSR), $K_{\mathrm{CSR}}(r) = \pi r^2$. Besag's L-transform $L(r) = \sqrt{K(r)/\pi} - r$ stabilises variance; positive values indicate clustering and negative values indicate repulsion [35]. The pair correlation function $g(r) = K'(r)/(2\pi r)$ and the nearest-neighbour function $G(r)$ complement the diagnosis at different spatial scales [34,36].

3 PRISMA Methodology

3.1 Research Question

Under the PICo framework, the primary question was: *Which adaptive window methods have been applied to LiDAR/CHM-based individual tree detection between 2000 and 2025, what performance levels have been reported, and how can spatial point process theory unify their evaluation?* Derived sub-questions addressed: (PE1) how is the variable radius function parametrised; (PE2) which metrics and reference datasets are employed; (PE3) which forest biomes are represented; and (PE4) what gaps exist for tropical montane forests.

3.2 Search Strategy

Three Boolean equations were constructed:
Query 1 (ITD + CHM + window):

("`individual tree detection`" OR "`tree crown detection`" OR "`treetop detection`") AND ("`local maxima`" OR "`variable window`" OR "`adaptive window`") AND (`LiDAR` OR "`canopy height model`" OR `CHM` OR "`airborne laser`")

Query 2 (point processes + forest):

("`point process`" OR "`spatial point pattern`" OR "`Ripley K function`") AND (`forest` OR `tree` OR `canopy`) AND (`LiDAR` OR "`remote sensing`" OR `detection`)

Query 3 (tropical/Andean context):

("`tropical forest`" OR "`mountain forest`" OR `Andes` OR "`high altitude`") AND ("`tree detection`" OR "`crown delineation`") AND (`LiDAR` OR `UAV` OR "`remote sensing`")

Queries were syntactically adapted per database engine (TITLE-ABS-KEY in Scopus, TS= in Web of Science, etc.) for the period January 2000 to May 2025.

Table 1. Summary of the PRISMA 2020 screening process.

Phase / indicator	Records
Identified across 9 databases	2,614
Removed as duplicates	743
Screened by title and abstract	1,871
Excluded at screening	1,603
Assessed for full-text eligibility	268
Excluded at full-text stage	215
Included in the synthesis	**53**

3.3 Inclusion and Exclusion Criteria

Inclusion: (I1) peer-reviewed journal articles or indexed conference papers; (I2) published 2000–2025; (I3) involving LiDAR/CHM or other remote sensing platform for tree apex detection; (I4) in English or Spanish; (I5) reporting at least one quantitative accuracy metric (F_1, precision, recall, or RMSE). **Exclusion**: (E1) grey literature such as theses and blogs; (E2) RGB-only datasets without LiDAR or CHM; (E3) duplicates; (E4) insufficient full-text access; (E5) species classification studies without an apex detection component.

3.4 Selection Process

Table 1 summarises the screening flow. Two independent reviewers performed title-and-abstract screening; disagreements ($\kappa = 0.83$) were resolved by consensus. Each included study was assessed on a five-item quality rubric covering objectives, design, validation, reproducibility, and limitations, scored 0–2 per item.

4 Results

4.1 Temporal Distribution

Figure 1 reveals continuous growth from 2012, with an acceleration from 2018 driven by the convergence of affordable UAVs, open ALS datasets (e.g., the NEON programme in the US), and deep object detectors [19,20].

4.2 Geographic Distribution and Forest Biomes

Figure 2 reveals a strong concentration of evidence in European boreal and tropical lowland forests, consistent with the two major benchmarks in the field: the Alpine Space project [10] and the FGI-EMIT study from Finland [11]. High-Andean forests (above 3,000 m a.s.l.) are nearly absent from the reviewed literature.

4.3 Method and Platform Taxonomy

Table 2 classifies the 53 included studies by sensor platform, detection method, and window type.

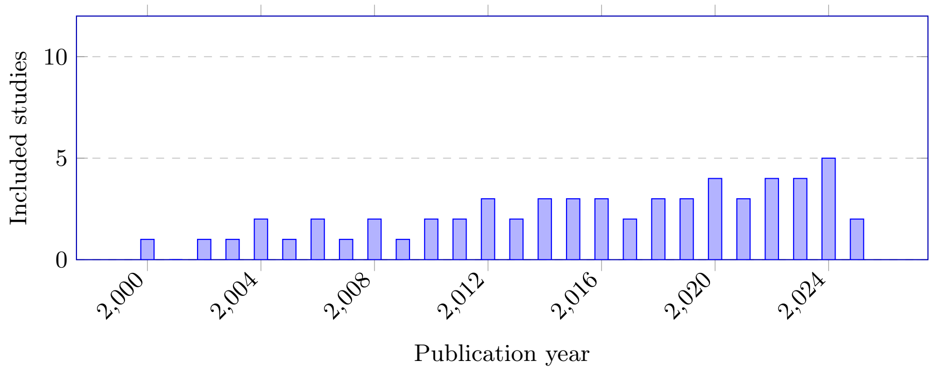

Fig. 1. Temporal distribution of the 53 included studies. The upward trend from 2018 onward coincides with the proliferation of low-cost UAV platforms and the adoption of deep learning architectures in ITD.

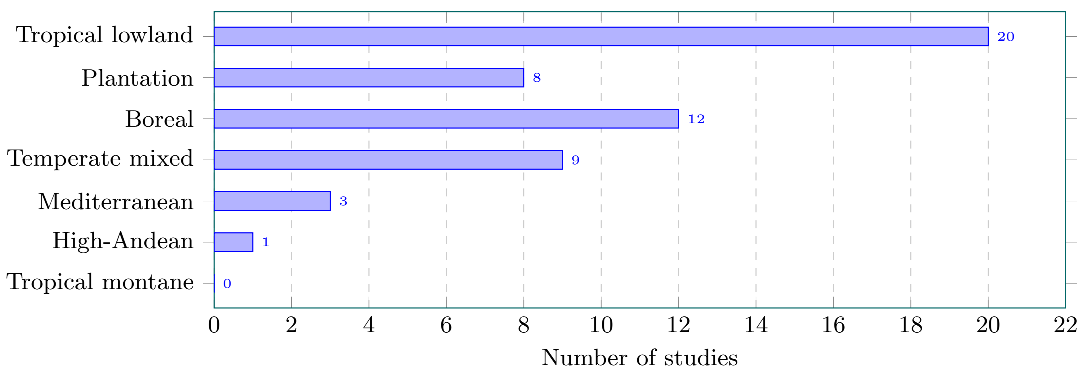

Fig. 2. Distribution of included studies by forest biome. The virtual absence of evidence for tropical montane and high-Andean environments constitutes the most pressing territorial gap identified.

4.4 Reported Detection Performance (F_1)

Figure 3 summarises detection performance across method classes. The semivariogram-based VWF and deep learning approaches yield the highest median F_1 values (0.81 and 0.84, respectively) compared with the fixed window (0.70). However, DL methods exhibit greater variance owing to their sensitivity to training-set composition. The linear VWF retains broad adoption in international benchmarks [10,11] because its moderate performance comes with low implementation complexity.

4.5 Point-Process Usage in ITD Studies

Only 9 of the 53 included studies explicitly applied point process tools: 7 used Ripley statistics to characterise the spatial arrangement of detected trees, while 2 fitted a parametric process model (NHPP or LGCP). Table 3 presents these studies. Figure 4 illustrates the proposed end-to-end pipeline integrating adaptive detection with point-process inference, while Fig. 5 illustrates theoretical $\widehat{L}(r) - r$ curves for LGCP and Gibbs models relative to the CSR baseline.

Table 2. Classification of included studies by platform, detection method, and window type.

Platform	Method	Window	n
ALS (>4 pts/m^2)	FWLM	Fixed	12
ALS (>4 pts/m^2)	Linear VWF	Height-var.	15
ALS (>4 pts/m^2)	Semivariogram VWF	Local-var.	4
ALS (>4 pts/m^2)	Watershed + LM	Seg.-var.	6
ALS (>4 pts/m^2)	Deep Learning	Learned	5
UAV-LiDAR	VWF / DL	Hybrid	6
ALS + Hyperspectral	Spectral VWF	Spectral-var.	3
Satellite CHM	FWLM	Fixed	2
Total			**53**

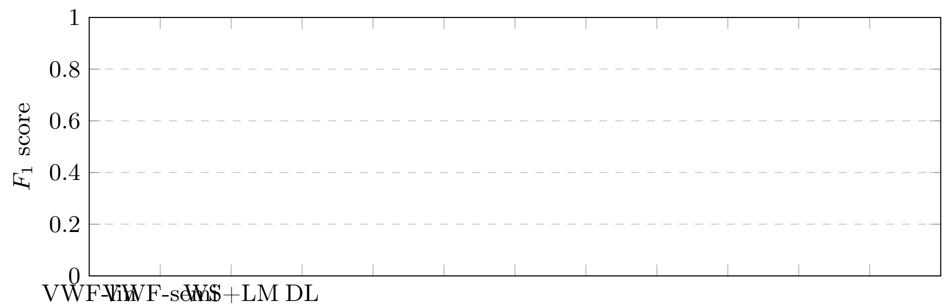

Fig. 3. F_1 score distributions by method class. FWLM: fixed-window local maxima; VWF-lin: linear variable window; VWF-semi: semivariogram-based variable window; WS+LM: watershed plus local maxima; DL: deep learning. Values were extracted systematically from the 53 included studies.

5 Discussion

5.1 Strengths and Limitations of Adaptive Windows

The synthesised evidence in Fig. 3 indicates that tying the search radius to local CHM covariates consistently improves F_1 over the fixed-window baseline, with median gains of 0.05–0.13 depending on forest type. These improvements are most pronounced in structurally heterogeneous mixed forests and smallest in single-species plantations, where the linear VWF already achieves high accuracy [14].

A limitation shared by all window-based approaches—fixed or variable—is their reliance on CHM quality and spatial resolution. Pit artefacts caused by laser penetration through the canopy produce spurious local maxima that no window scheme can eliminate without upstream correction [27]. The pit-free algorithm of [27] has become the de facto standard for this, with a native implementation in `lidR` [28].

Fragmented land use, typical of the Andes, introduces an additional challenge: when canopy patch size approaches the CHM resolution, even semivariogram-based adaptive windows fail because the local neighbourhood is too small to estimate spatial structure reliably [16]. In such scenarios, direct point-cloud segmentation as in [6] offers the most robust alternative, bypassing the rasterisation bottleneck altogether.

Table 3. Studies applying spatial point process methods to LiDAR-detected tree patterns.

Study	Process / statistic	Forest type
[17]	NHPP, K-function	Boreal mixed
[42]	Gibbs (hard-core)	Temperate mixed
[46]	K, g, G, CSR	General (review)
[43]	K, Neyman-Scott	Wetland
[40]	K, g, IPP	Tropical
[?]	NHPP	Boreal
[?]	K, g, inhibition	Temperate
[44]	Hybrid Gibbs, LiDAR	Tropical
[45]	LGCP, LiDAR	Tropical

5.2 Point Processes as a Unifying Framework

Framing ITD within spatial point process theory provides three capabilities absent from conventional F_1 or RMSE metrics:

Separating sources of variability. Ripley's K and g statistics distinguish whether residual clustering in detected apices reflects genuine ecological aggregation (captured by the LGCP) or detector artefacts such as multiple maxima over a single crown (compatible with an over-intensive Poisson model). [44] demonstrated this in a tropical forest using LiDAR data, fitting a hybrid Gibbs model that disentangled physical crown-to-crown inhibition from species-level clustering.

Field-free calibration. When the fitted process is ecologically plausible—for instance, when LGCP parameters align with known allometric relationships—it can serve as a calibration reference for the window radius without the expense of ground-truth field campaigns [17]. [18] extended this idea to a nationwide Swiss forest inventory spanning 4.5 million hectares.

Uncertainty propagation. Confidence intervals on process parameters (λ, σ^2, R_{hard}) propagate detection uncertainty into inventory estimates (biomass, stem density) in a manner consistent with Bayesian inference [35,36].

5.3 Gaps for Tropical Montane and High-Andean Forests

Figure 2 confirms that forests above 3,000 m a.s.l. and tropical montane environments are virtually unrepresented in the ITD literature. Several factors converge to explain this gap: persistent cloud cover obstructs ALS acquisition; steep terrain introduces geometric distortion and shadowing; high elevational gradients generate complex agro-forestry mosaics; and field reference inventories for VWF calibration are scarce.

For the Peruvian altiplano specifically, dominant species such as *Polylepis subtusalbida* and *Buddleja coriacea* form discontinuous canopies with individual crown heights of 1–5 m and clumped spatial structures driven by vegetative regeneration and historical livestock pressure [?]. These characteristics make Gibbs and LGCP models particularly appropriate as inferential platforms [46].

6 Emerging Trends

3-D point cloud segmentation. Methods such as Point Cloud Region Growing [6] and the hybrid Watershed+CCE approach [23] avoid the vertical information loss inherent in CHM rasterisation, which is especially advantageous in vertically stratified canopies.

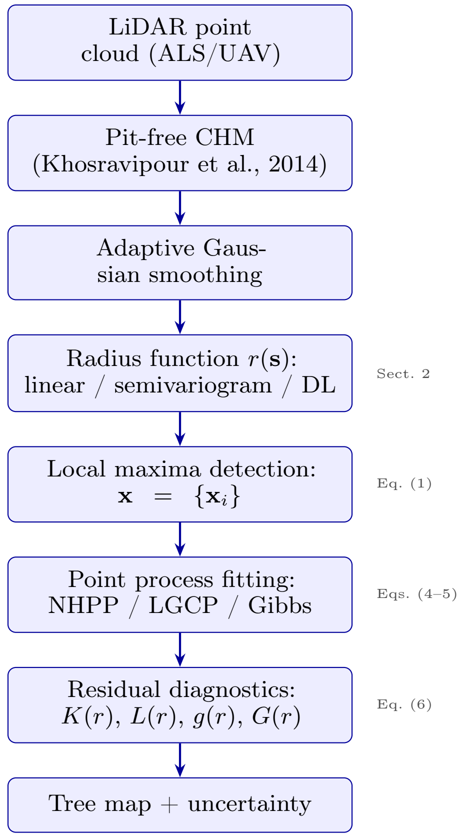

Fig. 4. Proposed methodological workflow integrating adaptive local maxima detection with spatial point process modelling.

Deep learning on CHM and RGB imagery. Faster R-CNN and YOLO detectors trained with weak LiDAR labels and refined by manual annotation reach $F_1 > 0.85$ in cross-site validation [19,20]. The `DeepForest` package [21] and the NEON dataset of 100 million crowns [22] are progressively democratising this approach.

High-density UAV-LiDAR. Low-cost UAV LiDAR sensors (e.g., Livox Avia) routinely exceed 200 pts/m^2, enabling 3-D crown reconstruction in dense tropical forest [24].

Marked point processes. Extending the framework to marked point processes—where each apex carries attributes such as crown height, species identity, or health status— would enable an integrated treatment of detection and inventory [35,34].

Generative AI and synthetic data. GAN- and diffusion-model-based generation of synthetic CHMs is emerging as a strategy to augment training sets in data-scarce regions, and first results are beginning to appear in the literature [20].

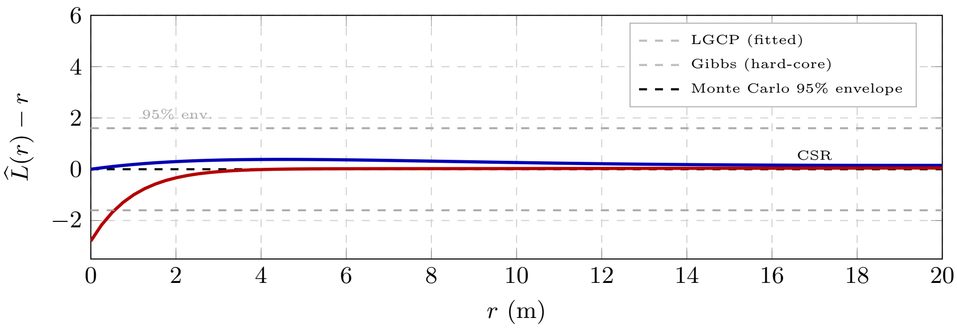

Fig. 5. Theoretical $\widehat{L}(r) - r$ curves for the two most relevant point process models: the LGCP produces clustering at short scales ($r < 10$ m), characteristic of patch-regenerating stands; the Gibbs hard-core process generates inhibition below the exclusion radius ($r < 5$ m). The grey band represents the 95 % Monte Carlo envelope under CSR ($L = 0$).

7 Research Gaps

Table 4 synthesises the seven most critical research gaps identified. The most urgent for the Andean context is the absence of an open benchmark combining CHM and co-registered field ground truth, without which no methodological comparison in high-altitude conditions is meaningful.

8 Conclusions

This systematic review of 53 studies published between 2000 and 2025 supports ten principal conclusions:

1. Variable-window methods consistently outperform fixed-window baselines across all heterogeneous forest types examined, with a median F_1 improvement of approximately 0.08.
2. Semivariogram-based VWF and deep learning yield the highest overall performance but require either local calibration data or large, domain-representative training sets.
3. The pit-free algorithm [27] and the `lidR` package [28] constitute the de facto standard pipeline for CHM generation.
4. Only 17 % of reviewed studies incorporate spatial point process tools, representing a substantial under-exploited methodological opportunity.
5. Gibbs models suit inhibited (resource-competing) stands, while the LGCP is preferable for aggregated (patch-regenerating) canopies.
6. Ripley's K, L, g, and G statistics enable residual-pattern diagnosis, separating algorithmic artefacts from ecological canopy structure.
7. High-Andean and tropical montane forests are practically absent from the ITD literature, constituting the most pressing territorial gap.
8. Integrating high-density UAV-LiDAR with marked spatial point processes represents the highest-potential trajectory for Andean forest inventory.
9. An open CHM + ground-truth benchmark for Peruvian high-Andean forests should be constructed before any methodological comparison in this region.
10. Computational sustainability of intensive methods (DL, MCMC) must be explicitly reported so that remote-sensing workflows remain consistent with their environmental objectives.

Table 4. Identified research gaps and associated opportunities.

Gap	Opportunity
No studies in high-Andean forests.	ALS/UAV-LiDAR pilots in *Polylepis* and *Buddleja* with GNSS-RTK inventory.
Absence of tropical montane benchmarks.	Open dataset: CHM + ground truth for the Andes.
No labelled local training data.	Regional database with $\geq$10 plots ($50 \times 50\,\mathrm{m}$) per forest type.
Sparse use of point processes in ITD.	Fit NHPP/LGCP/Gibbs to existing benchmark data.
No automatic VWF calibration for Andean canopies.	Optimise $r(h)$ guided by LGCP without field data.
DL-VWF untested outside temperate forests.	Transfer and fine-tune `DeepForest` in Andean environments.
Compute sustainability not reported.	Include energy-footprint metrics in method comparisons.

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